

OpenDRS User's Guide

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Introduction

The Open-access Dietary Recall System is a 24-hour dietary recall programmed in XLSForm language for data collection on Android tablets. This program is provided as an open-source template for use in not-for-profit nutrition research projects in low- and middle-income countries. A demonstration of the template OpenDRS 24-hour recall may be viewed at:

https://maziratest.surveyccto.com/collect/opensdrs_demo_surveyccto?caseid=

Users may modify the tool as needed to support dietary data collection. The following sections of this User's Guide provide a brief overview of the steps required to adapt the tool to a survey setting and to calculate observed nutrient intakes from the data collected.

Use Agreement for OpenDRS

OpenDRS is provided free and open-source. Download or use of any OpenDRS components or files is considered acknowledgement of the following terms:

1. Use of OpenDRS is limited to not-for-profit research or surveys.
2. Any publication, report or presentation based on data collected using OpenDRS should include a citation acknowledging the developers of the tool. See the OpenDRS OSF page (<https://osf.io/j3axd/>) for example citations.
3. Users of OpenDRS take full responsibility for all aspects of use within their projects. This includes, but is not limited to: adaptation of the tool to the research setting, piloting, staff training, data collection activities, data quality control and data processing and analysis. The authors of OpenDRS have provided a template and are not responsible for how users apply that template. OpenDRS should be used by researchers with prior experience and/or advanced training in detailed dietary assessment.

OpenDRS files provided on Open Science Framework

The following files are provided on the Open Science Framework at <https://osf.io/j3axd/> :

- **OpenDRS.User Guide_20190603.pdf** An overview of how to adapt and implement OpenDRS for collection of 24-hour dietary recalls in a particular research or survey setting.
- **OpenDRS_Demo_ODK.xlsx** A template 24-hour recall for use with ODK. This version does not include household recipe data collection.
- **OpenDRS_Demo_ODK.xml** The .xml version of the OpenDRS ODK template without recipe data collection.
- **OpenDRS_Demo_SurveyCTO.xlsx** A template 24-hour recall for use with SurveyCTO. This version does not include household recipe data collection. The program functions in the same way as the ODK version, but one command and the formatting of translation column headings vary between ODK and SurveyCTO.
- **OpenDRS_Demo_SurveyCTO.xml** The .xml version of the OpenDRS SurveyCTO template without recipe data collection.
- **OpenDRS Demo Attachments** Folder containing five required attachments to the ODK or SurveyCTO demos, including food and ingredient lists (foodlist.csv and ingredients.csv) and food identification photos (MediumPorridge.jpg, ThickPorridge.jpg and ThinPorridge.jpg).

- **form6_long_v6.do** Sample Stata code for converting OpenDRS data from wide format (one row per recall) to long format (one row per reported food or drink).

Contact

The primary author and contact for OpenDRS is Bess Caswell (blcaswell@ucdavis.edu). Dr. Caswell is a postdoctoral scholar in the Program in International and Community Nutrition at the University of California, Davis. Please contact her about your project using OpenDRS. User questions and feedback will be used to update this document and the OpenDRS tools.

Structure of the 24-hour Recall

The OpenDRS template is intended to replicate the interactive, multiple-pass dietary recall described by Gibson and Ferguson (2008) for use in low- and middle-income countries. The recall is organized into the following passes:

- **Pass 1: Quick listing of all foods and drinks consumed during the previous day**
In this pass, the interviewer asks the respondent to list all foods and drinks consumed from waking the previous day to waking the day of the recall. Sample prompts are shown to remind the interviewer of probes to use to help the respondent recall the entire day and any episodes when food or drink was consumed. A very short description, using the respondents own terms for the food or drink, is recorded in a text field. After each recorded item, the interviewer is prompted by a pop-up window to indicate whether there are more foods or drinks to record ('Add another group?'). When the respondent cannot recall any additional foods or drinks, the interviewer selects 'Do not add' when prompted, and the recall proceeds to Pass 2.
- **Pass 2: Collection of detailed description of each food or drink and factors such as place of preparation and time consumed**
Passes 2, 3 and 4 will be repeated one time for each food or drink reported in Pass 1. In Pass 2, a series of increasingly specific food descriptions is shown to the interviewer. The interviewer asks the respondent to provide the needed level of detail to distinguish among the food groups or individual food items shown, until a final, specific food item is selected. The program then prompts the interviewer to ask the respondent which ingredients were used, if the food was a mixed dish. Likely ingredients based on local cuisine are displayed, or the interviewer can type in other ingredients used. The food selection questions and ingredient lists are set by the user in external .csv files attached to the recall program. The time of consumption is also recorded in Pass 2, and other questions, such as place of preparation, may be added.
- **Pass 3: Estimation of portion size of food or drink consumed, and collection of recipe data, as applicable**
In Pass 3, the interviewer asks the respondent to demonstrate the amount of food or drink consumed. If household recipe data is collected, this is included in Pass 3.
- **Pass 4: Review of recorded foods and drinks and final probe for additional items**
In Pass 4, all recorded details for each food or drink are displayed for the interviewer to check and review with the respondent. If corrections are needed, these can be entered in a text field. If additional foods were recalled after Pass 1 was completed, these can be entered in Pass 4. For each additional food, all of the questions from Passes 1-4 are repeated.
- **Questions on appetite and supplements**

Following the four recall passes, a series of questions is presented to collect information on whether food intake may have been unusual on the recall day or whether any supplements were taken in the past week.

Selecting a Survey Platform

OpenDRS is written in XLSForm and will only operate on survey platforms which use the same language. These platforms include Open Data Kit (ODK, <https://opendatakit.org>), SurveyCTO (<https://surveycto.com>) and others. See the XLSForm site (<http://xlsform.org>) for further options.

Users are strongly cautioned to test OpenDRS via a free trial or demonstration before committing to a survey platform. XLSForm compilers and survey design software vary among platforms. OpenDRS may require detailed modifications to run on platforms using propriety software that compiles XLSForm differently than ODK or SurveyCTO.

Users are advised to shop carefully when choosing a survey platform, as costs and features vary. Exploration of features through free trials or demonstration sites is recommended. Some of the factors users should consider are:

- Data management and storage features
 - Some platforms will require substantial in-house data management and storage resources, including IT equipment, staff time and expertise
 - Fee-based services may provide a range of these features within the subscription package
- Technical support
 - Subscription services may include direct assistance
 - The depth and quality of online documentation may vary among platforms
- Quality and ease of user interface for the researcher and data collector
- Data security
 - Review whether data is encrypted during transmission, storage on the data collection devices and storage of compiled datasets on a server.
 - Varying degrees and forms of data encryption may be provided by subscription services or developed and maintained by research staff.
 - Check data security requirements (such as IRB or Good Clinical Practice) against in-house IT resources and subscription service features.
- Survey implementation features vary among platforms. Users should consider whether any of the following are needed:
 - Connectivity options for data collection
 - Mobile device connected to cellular data signal or wifi
 - Mobile device operating offline
 - Web-based data collection
 - Case management tools, such as linked records or visit scheduling
 - Recruitment emails
 - Online / real-time data monitoring tools
 - Programming flexibility:
 - Online, drag-and-drop form creation vs. writing code in XLSForm (or both)

- Complex survey structures
- Offline form development capability (how much internet connectivity is required for writing, compiling and testing forms?)

Customizing the Food List and Ingredient Files

The food list and ingredient files are the critical piece of customization in the OpenDRS system. Each of these is an external .csv file that is developed to reflect the diet of the study population. Each needs to be as complete and comprehensive as possible, though they can be updated at any time during the study without changing the OpenDRS form definition. If data from previous 24-hour dietary recalls in the study area are available, these are an ideal resource for developing comprehensive food and ingredient lists. In absence of such data, or if the data is outdated, users will need to compile a thorough listing of foods consumed in the study area and a detailed picture of how those foods are consumed and prepared. This may be through drawing on prior studies, local food composition tables, local or regional cookbooks, interviews and listing activities with local staff and community members, market surveys, a quick-list dietary recall survey of the study area, or other activities and resources. Though the food and ingredient lists can be updated during the study, establishing thorough initial lists with foods and ingredients well organized by groups and sub-groups will set a solid foundation that can be refined over time. Several weeks of researcher time should be budgeted for this activity, especially if prior 24-hour recall data are unavailable or outdated.

Use of the food list within the OpenDRS form

The food list is referenced in Pass 2 of the OpenDRS form to describe foods or drinks consumed with a level of detail that they can be linked to recipe and food composition data. The detailed food identification process is executed as a series of increasingly specific food groupings, ending in the selection of a specific food item in the food list. Though XLSForm can support searchable or scrollable lists, this format was chosen to reduce data entry errors, such as inadvertent selection of a response while scrolling or missing an available food when skimming many responses or using a different spelling. Users may wish to test whether a search field or scrollable list is a better choice for their research setting. In the authors' field experience, interviewers quickly learn how foods are organized by group and are able to very rapidly navigate through the food group questions to arrive at a specific food item. If the interviewer has located the correct sub-

Figure 1. Example of Pass 2 questions to collect detailed food identification

The figure displays three sequential screenshots of the OpenDRS form during Pass 2 of a 24-hour dietary recall. The first screenshot, titled 'Collect > Form 6: 24-hour Recall', shows a list of food groups under the heading 'SELECT GROUP CORRESPONDING TO Porridge'. The groups include Beans / peas, Bread, rice or pasta, Drink, Fish, Fruit, Meat, egg or insect, Nsima, Porridge (selected), Snack or sweet, Vegetable or vegetable relish, Soup, and Other foods. The second screenshot shows the 'SELECT GROUP CORRESPONDING TO Porridge' screen with three images of porridge. The first image is labeled 'Thin consistency', the second 'Medium consistency' (selected), and the third 'Thick consistency'. The third screenshot shows the 'SELECT FOOD' screen with a list of specific food items. The items include Infant cereal, Cerelac or Purity, medium consistency; Phala, grainmill maize, medium consistency (selected); Phala, likuni, medium consistency; Phala, mgaiwa, medium consistency; Phala, rice, medium consistency; Phala, soya flour, medium consistency; Phala, woyera, medium consistency; and Other phala, medium consistency.

group and the item is not listed, the interviewer may select the 'other' response within the sub-group and type in a detailed description of the food.

Compiling the food list

The food list should contain all commonly consumed foods and drinks that may be reported by the respondents in the 24-hour recall. The food list should reflect foods as consumed in the target population, including mixed dishes, commercially-produced or home-prepared drinks, foods consumed as an unmixed item (fruits, commercially-produced snacks) and other foods. If household recipe data collection will be included, all common ingredients should also be in the food list.

After assembling a list of foods and drinks, all of the items must be sorted into food groups and sub-groups, to arrange the foods into a selection framework. It is recommended to sort food groups and sub-groups according to local food consumption patterns that will be intuitive to interviewers. To prevent scrolling, the number of items in any food group or sub-group should be limited to the number of response options that can fit on a tablet screen. This can be tested by creating a simple form with a single `select_one` field that contains at least twenty response options. This can then be loaded onto a tablet that will be used for data collection, so that the number of response options that show without scrolling can be counted.

Codes must be assigned to each specific food or drink and to each food group and sub-group. Coding should be planned carefully to allow items to be added and avoid ambiguity. Specific food item codes may be matched to food composition table or recipe database codes if desired. If the food list is modified during piloting or over the course of the study, users are strongly advised to assign additional codes rather than changing existing codes, as changing codes may result in later analytic errors.

For an example of how the food group classification should be organized, please see the example file, `foodlist.csv`, in the 'OpenDRS Demo Attachments.zip' folder, among the files on the OpenDRS OSF page (<https://osf.io/j3axd/>). If this format is not followed, the selection framework will not function correctly. For further reference, see [Resources for learning XLSForm](#) below, particularly <http://xlsform.org/en/#pre-loading-csv-data>.

An 'other' item should be included within each sub-group listing of specific food items, and the descriptor must start with the word 'Other.' Selection of a specific food starting with the word 'Other' will trigger a text prompt for the interviewer to type in a description of the food which was not found in the preset food list. An 'Other' category should also be included in the first set of food groups, so that interviewers can access this text prompt for foods that don't fit within existing food groups or sub-groups. Changes to this system will require modification of the form definition, or the text prompt for interviewers to record foods or drinks not on the food list may not function properly.

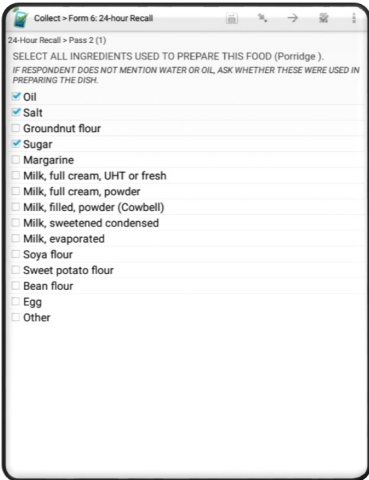
The food list file must be saved using the file name 'foodlist.csv' or the OpenDRS form definition must be updated wherever this file name is referenced. Similarly, column names should be retained from the template `foodlist.csv` or updated in the form definition.

The OpenDRS form definition and sample foodlist assumes that one main food group and one sub-group is sufficient to filter down to a list of specific food items that will fit on one screen. If additional sub-group levels are needed, both the form definition and `foodlist.csv` will need to be modified accordingly.

Use of the ingredient list within the OpenDRS form

The ingredient list is referenced in Pass 2 of the OpenDRS form to populate response options for a select multiple question recording the ingredients that were used in a mixed dish. This ingredient data can then be used to link the item to a recipes database. If household recipes will be collected, the ingredient question in Pass 2 could be removed (and the ingredient.csv attachment eliminated), or the ingredient question in Pass 2 may be used as an initial quick list of ingredients to collect details on in Pass 3.

Figure 2. Example of Pass 2 question to record ingredients in a mixed dish



The screenshot shows a mobile application interface for a survey. At the top, it says 'Collect - Form 6: 24-hour Recall'. Below that, the question is '24-Hour Recall - Pass 2 (1)' and 'SELECT ALL INGREDIENTS USED TO PREPARE THIS FOOD (Porridge)'. A note states: 'IF RESPONDENT DOES NOT MENTION WATER OR OIL, ASK WHETHER THESE WERE USED IN PREPARING THE DISH.' The list of ingredients includes: Oil (checked), Salt (checked), Groundnut flour (unchecked), Sugar (checked), Margarine (unchecked), Milk, full cream, UHT or fresh (unchecked), Milk, full cream, powder (unchecked), Milk, filled, powder (Cowbell) (unchecked), Milk, sweetened condensed (unchecked), Milk, evaporated (unchecked), Soya flour (unchecked), Sweet potato flour (unchecked), Bean flour (unchecked), Egg (unchecked), and Other (unchecked).

Compiling the ingredient list

The ingredient list should contain ingredients that are typically used to prepare mixed dishes in the target population, excluding main ingredients that may be used to specifically identify the mixed dish within the food list. Mixed dishes should be grouped according to typical preparation style. For each mixed dish type, a list of common ingredients should be compiled. If an ingredient is commonly used in more than one type of mixed dish, it should be repeated in the ingredients list for each type of mixed dish where it may be used.

For an example of how the ingredients list should be organized, please see the example file, ingredients.csv, in the 'OpenDRS Demo Attachments.zip' folder, among the files on the OpenDRS OSF page (<https://osf.io/j3axd/>). If this format is not followed, the pre-set ingredient lists by type of mixed dish will not function correctly.

The ingredients file must be saved using the file name 'ingredients.csv' or the OpenDRS form definition must be updated wherever this file name is referenced. Similarly, column names should be retained from the template ingredients.csv or updated in the form definition.

The OpenDRS form definition assumes that no more than fifteen ingredients will be selected for any mixed food. If it is possible that more than fifteen ingredients may be listed for one mixed dish, the form definition and foodlist.csv will need to be modified. As with the food list, it is recommended to limit the number of response options in each ingredient list to the number that will fit on the screen without scrolling.

Customizing the Recall and Supporting Questions

The OpenDRS form definition (.xlsx file) may be adapted as needed to support the needs of a given project. Such modification will require the user to become familiar with XLSForm programming language. Substantial online documentation for learning how to use XLSForm can be found at the sites listed below.

Resources for learning XLSForm

- <http://xlsform.org>
- <https://docs.opendatakit.org/> See pages listed under 'Form Building' in the navigation list (black bar on the left)
- <https://forum.opendatakit.org/> ODK community site where users can search previously posted questions and answers or post their own question
- Subscription-based sites offer additional online documentation on how to use XLSForm. This may be accessible with sign-up for a free trial or may require paid subscription.

Customizing the Recall

Once a user is familiar with XLSForm, the OpenDRS template can be modified to customize elements of the 24-hour recall protocol. Simple modifications include altering question labels or response options, adding or removing requirement that question be answered, changing question constraints, or changing question relevance (skip logic).

Use caution when modifying questions, as changes may affect conditional logic required for later portions of the recall. When modifying a question, use the 'Find' tool in Excel to search for any later instances of the field name. If the name of the field being modified appears in the relevance criteria, constraint definition or label of later questions, these will need to be updated to reflect changes made to the modified question.

Example of altering question labels and response options

In the template, portion size is estimated in mL using dry rice, modeling dough or water as a model, and household or locally purchased dishes as memory aids. These questions appear as shown in Figure 1.

Figure 3. Portion size estimation questions in OpenDRS template

The figure displays three sequential screenshots of the OpenDRS 24-hour recall form, specifically the '24-Hour Recall - Pass 3 (1)' section. The first screenshot shows a question: 'SELECT MATERIAL USED TO ESTIMATE PORTION SIZE FOR Porridge (Phala, grainmill maize, medium consistency; Mid-day (11 am - 2 pm))'. Below the question are four radio button options: 'RICE' (selected), 'WATER', 'DOUGH', and 'CANNOT REPORT'. The second screenshot shows a question: 'SELECT UTENSIL USED TO ESTIMATE PORTION SIZE FOR Porridge (Phala, grainmill maize, medium consistency; Mid-day (11 am - 2 pm))'. Below the question is a list of utensils with radio button options: 'RECALL BOWL', 'OTHER SMALL BOWL', 'MEDIUM BOWL', 'LARGE BOWL', 'SMALL PLATE', 'LARGE PLATE', 'SMALL CUP', 'LARGE CUP', 'SMALL SPOON', 'LARGE SPOON', and 'OTHER, SPECIFY'. The third screenshot shows a question: 'ENTER PORTION SIZE IN MILLILITERS FOR Porridge (Phala, grainmill maize, medium consistency; Mid-day (11 am - 2 pm))'. Below the question is a text input field containing the number '55'. At the bottom of the form is a numeric keypad with buttons for digits 1-9, 0, a decimal point, and a backspace key.

These three questions appear in the 'survey' tab of the OpenDRS form definition as:

select_one method	drportionmthd	SELECT MATERIAL USED TO ESTIMATE PORTION SIZE FOR \${fooddesc4pass3} (\${spec4pass3}; \${time4pass3})
select_one utensil	drportionutensil	SELECT UTENSIL USED TO ESTIMATE PORTION SIZE FOR \${fooddesc4pass3} (\${spec4pass3}; \${time4pass3})
integer	drportion	ENTER PORTION SIZE IN MILLILITERS FOR \${fooddesc4pass3} (\${spec4pass3}), \${time4pass3})

The associated response options appear in the 'options' tab of the OpenDRS form definition as:

method	1	RICE
method	2	WATER
method	3	DOUGH
method	99	CANNOT REPORT
utensil	1	RECALL BOWL
utensil	2	OTHER SMALL BOWL
utensil	3	MEDIUM BOWL
utensil	4	LARGE BOWL
utensil	5	SMALL PLATE
utensil	6	LARGE PLATE
utensil	7	SMALL CUP
utensil	8	LARGE CUP
utensil	9	SMALL SPOON
utensil	10	LARGE SPOON
utensil	88	OTHER, SPECIFY

To change these questions to measure portion size in grams using foods available at the survey location and a food scale or food models labeled with the equivalent weight in grams, the 'survey' and 'choices' tabs can be updated as highlight in red text below:

select_one method	drportionmthd	SELECT METHOD USED TO ESTIMATE PORTION SIZE FOR \${fooddesc4pass3} (\${spec4pass3}; \${time4pass3})
select_one utensil	drportionutensil	SELECT MATERIAL USED TO ESTIMATE PORTION SIZE FOR \${fooddesc4pass3} (\${spec4pass3}; \${time4pass3})
integer	drportion	ENTER PORTION SIZE IN GRAMS FOR \${fooddesc4pass3} (\${spec4pass3}), \${time4pass3})

method	1	WEIGHT BY FOOD SCALE
method	2	FOOD MODEL
method	3	DOUGH
method	99	CANNOT REPORT
utensil	1	GRAMS OF FOOD
utensil	2	GRAMS OF MODEL RICE
utensil	3	GRAMS OF MODEL BREAD SLICE
utensil	4	GRAMS OF WATER
utensil	5	SMALL PLATE
utensil	6	LARGE PLATE
utensil	7	SMALL CUP
utensil	8	LARGE CUP

utensil	9	SMALL SPOON
utensil	10	LARGE SPOON
utensil	88	OTHER, SPECIFY

Adding questions

Questions may be added before, within or after the 24-hour recall. To add a question, insert a new row in the OpenDRS form definition and define the question according to guidance from the XLSForm references listed above. Take care to note the location of ‘begin group’ and ‘end group’ (or ‘begin repeat’ and ‘end repeat’) rows in the form. If the question is inserted within a group or repeat, it will be bundled together with the other questions in that group. This can determine whether the question appears on the same screen as other questions, whether group relevance criteria are applied, or whether the question is repeated. To add questions before or after the recall, place them either before begin group: recall line (Row 17 in the OpenDRS template) or after the end group: recall line (Row 207).

Changing the structure of the recall

The multiple-pass structure of the recall can be modified by users who are well familiar with XLSForm language and programming logic. Before altering the structure of the recall, users should be familiar with using calculate fields, repeat loops, and linking data within and between repeat loops (including expressions such as jr:choice-name(), pulldata(), indexedrepeat(), position(..), etc.).

Compiling and Deploying the 24-hour Recall Form

Once any modifications have been made to the recall and supporting questions and the food list and ingredient files have been created for the study setting, the .xlsx form definition must be run through a compiler which converts the form to .xml. This .xml form, the .csv files containing the food and ingredient lists, and any photos or other media can then be deployed to Android devices for data collection. Deployment will typically involve installation of the survey platform’s data collection app onto the tablets and loading the compiled form definition with attachments onto the tablets. Depending on the platform and tablet hardware, this may be done over internet connection, USB or SD card file transfer or local wifi network. Users should refer to survey platform documentation for detailed instructions and options.

Any time the form definition, food or ingredient lists or media attachments are revised, the form and attachments should be updated on all tablets used to conduct 24-hour recall interviews. This can be done by repeating the process of downloading the form definition with attachments to each device.

Piloting

After deployment of the OpenDRS form to tablets, the form will need to be thoroughly piloted to check for any errors, including incorrect skip logic, misstated relevance or constraint criteria, non-functioning data links, missed or incorrect translations, omitted or unclear items in food and ingredient lists, etc. Piloting should include thorough and systematic checks by multiple users as well as test interviews. Piloting may be combined with the training process, so long as staff are kept informed of the piloting and revision process and any changes made.

Training Field Staff

Field staff conducting 24-hour recalls will need ample time to train on the use of the tablet app for data collection, correct 24-hour recall interview technique and use of the OpenDRS form. At least one full week of training is recommended. It is not a fixed requirement that interviewers have prior experience with Android devices or with dietary assessment, but if members of the field team do not have such experience, additional training time will be needed. In previous projects, staff with secondary school diplomas and little to no prior experience with Android devices or nutrition surveys have succeeded in learning and correctly implementing the OpenDRS 24-hour recall. The technical monograph by Gibson and Ferguson (link in References section) is recommended as a detailed guide on 24-hour recall implementation, correct interview procedure and staff training program.

The OpenDRS 24-hour recall training should include:

- Lessons on the purpose and method of the 24-hour recall
- Review of detailed Standard Operating Procedures or other document describing implementation of the 24-hour recall interview from start to finish
- Walk-through of the OpenDRS tablet form, including demonstration of any skip logic patterns
- Individual practice navigating the form and entering data, including
 - Drills to locate foods within the Pass 2 food identification framework
 - Drills to practice portion size estimation methods
- Mock interviews between interviewers
- Practice or pilot interviews with community members
- Review of tablet use policies and procedures for uploading data

Collecting and Uploading Data

Based on the research or survey setting and survey platform capabilities, OpenDRS can be used to collect data on offline Android tablets, cellular- or wifi-connected Android tablets or web browser. The specific methods for collecting data vary by survey platform. Survey platforms also vary in capacity to support different methods of uploading data from the tablets to a central server and back-up storage. Options for this data transfer and compilation include: upload by cellular or internet connection directly to a cloud server; local area network transfer to a central laptop for later upload to a physical or cloud server; USB transfer to a laptop or physical server. Please see the [Selecting a Survey Platform](#) section above for tips on exploring which platforms will be best suited for data collection and compilation in a given setting.

Analyzing the 24-hour Recall Data

Transposing the data from wide to long

Initial output of the OpenDRS data from the compiled data storage contains one row per 24-hour recall interviewer. For ease of analysis, the dataset should be transposed from this wide format to a long format with one row per food or drink recorded. If collecting household recipes, these may be best split off to a separate dataset and transposed separately to a long format of one row per ingredient. Please see sample Stata code for transposing OpenDRS data without household recipes from wide to long in the file `form6_long_v6.do` on the OpenDRS OSF site (<https://osf.io/j3axd/>).

Calculating nutrient intakes

Calculation of observed nutrient intakes should be carried out by a user with expertise in dietary assessment and statistical software. The following steps must be followed to calculate nutrient intakes from the long format OpenDRS data:

1. If portion sizes were not reported in grams of the food or drink item, they must be converted to grams using portion size conversion factors.
2. Mixed dishes must be disaggregated into ingredients using recipe data or linked to food composition values for the specific recipe. Choice between these two methods will depend on other objectives of the analysis, such as how dietary diversity or sources of key nutrients might be described.
3. Recipe ingredients and individual (unmixed) foods must be linked to food composition data. The nutrient contents of each ingredient or food can then be calculated by multiplying the consumed weight in grams by the nutrient composition values.
4. Energy and nutrient intakes from all individual foods and ingredients reported in each recall to obtain the observed energy and nutrient intakes during the recalled day.

Compiling supporting data tables

As with any 24-hour dietary recall, compilation of appropriate supporting data is critical. Careful planning and budgeting of researcher time are recommended for these steps.

Portion Size Conversion Factors

A table of portion size conversion factors must be compiled to convert portion sizes from the recorded units of measurement to grams. For example, in the OpenDRS template, portion sizes are recorded in mL. A portion size conversion value (the density of the food in g/mL) is associated with every item in the food list to convert the portion size from mL to g.

Conversion values may be collected by taking repeat weight and volume measures of foods local to the study area, or values may be found in references such as:

- The FAO / INFOODS Databases Density Database Version 2 . 0
<http://www.fao.org/infoods/infoods/tables-and-databases/faoinfoods-databases/en/>
- USDA National Nutrient Database for Standard Reference, Legacy edition
<https://ndb.nal.usda.gov/ndb/>
- University of Minnesota Nutrition Data System for Research (NDSR) density tables
<http://www.ncc.umn.edu/ndsr-database-page/>

Recipes Database

A recipes database must be compiled to describe the ingredients in each mixed dish and their relative quantities. Data for these recipes may be collected from household recipes (if including in the 24-hour recall), focus group or weighed observation data, recipes from a previous dietary survey, or other sources. The technical monograph by Gibson and Ferguson is recommended as a resource on compiling recipe data (link in [References](#) section).

Food Composition Table

A food composition table with items matching all foods, drinks or ingredients captured in the 24-hour recall table must be compiled. Advanced planning of this step may also aid in development of the food list, as availability of food composition values may guide the level of distinction needed between similar foods. The

food composition table should contain values – referenced, measured or imputed – for each nutrient of interest and each food in the table. FAO/INFOODS is a recommended resource for guidance on compilation of a food composition table and locating existing national and regional tables:

<http://www.fao.org/infoods/infoods/en/>

References

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